

0.1 Section 7.4 - Expected Value

Example 1

Roulette is a game where a ball is rolled on a circular track. There are 38 numbers in the track, 18 red, 18 black, 2 green. If you correctly guess the number the ball lands on, you win.

1. What is the probability that you win?

$$\boxed{\frac{1}{38}}$$

2. You play 10 times. Find the probability of winning 6 out of the 10 times.

This is just binomial probability

$$\binom{10}{6} \left(\frac{1}{38}\right)^6 \left(\frac{37}{38}\right)^4 \approx \boxed{0.0000006}$$

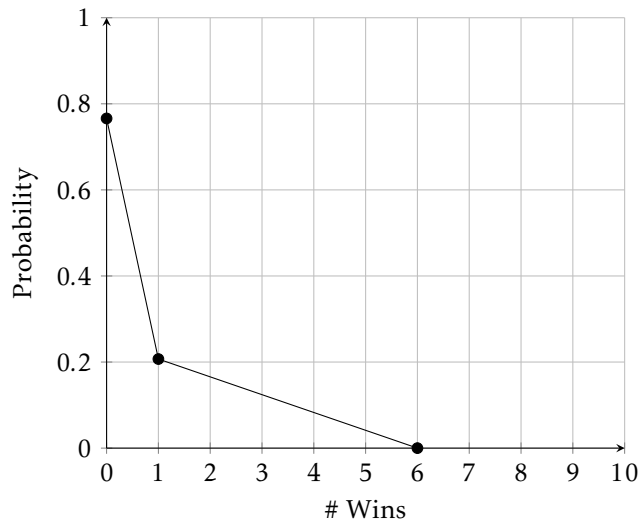
3. You play 10 times. Find the probability you never win

$$\left(\frac{37}{38}\right)^{10} \approx \boxed{0.766}$$

4. You play 10 times. Find the probability you win exactly once

$$\binom{10}{1} \left(\frac{1}{38}\right)^1 \left(\frac{37}{38}\right)^9 \approx \boxed{0.207}$$

Notice that if you graph the probability and wins, you get



5. You play 10 times. Find the probability of winning at least once

Let's use the compliment

$$\begin{aligned} P(\text{win at least once}) &= 1 - P(\text{never win in 10 spins}) \\ &= 1 - \left(\frac{37}{38}\right)^{10} \\ &\approx \boxed{0.234} \end{aligned}$$

In addition to guessing a number, you can also make a bet. The way it works is:

1. If you guess # correctly, then you win 35× your bet
2. If you guess incorrectly, then you lose your bet

Now we ask, how much do we win (or lose) **on average** when making a bet?

Assume we play a bunch of times. We bet \$1 every time

Let's play 38 times

Since the probability of winning is $\frac{1}{38}$, you'll probably win 1 time and lose 37 times

So

$$\begin{aligned} \frac{\text{total win/loss}}{\# \text{ played}} &= \frac{\overbrace{1}^{\# \text{ Wins}} \cdot \overbrace{35}^{\$ \text{ gained}} + \overbrace{37}^{\# \text{ losses}} \cdot \overbrace{(-1)}^{\$ \text{ lost}}}{\underbrace{38}_{\text{played 38 times}}} \\ &= \boxed{-0.05} \end{aligned}$$

Notice that we can simplify this into

$$\underbrace{\frac{1}{38}}_{P(\text{Win})} \overbrace{(35)}^{\$ \text{ Gained}} + \underbrace{\frac{37}{38}}_{P(\text{Loss})} \overbrace{(-1)}^{\$ \text{ Loss}}$$

This is called **expected value**

Definition 1 (Expected Value)

Expected Value is the what we expect to get from a probability outcome on average.

$$EV = \sum (\text{probability})(\text{payoff})$$

For every case

Example 2

Let's go back to the roulette example

We can also bet on red or black. If we choose correctly, we get paid the size of our bet. If we lose, we lose our bet.

What is the expected value of betting \$1 on red?

There are two cases.

1. We guess correctly, $P(\text{red}) = \frac{18}{38}$
2. We guess incorrectly, $P(\text{the rest}) = \frac{20}{38}$

$$\begin{aligned} EV &= \sum (\text{probability})(\text{payoff}) \\ &= P(\text{red}) \cdot (1) + P(\text{the rest}) \cdot (-1) \\ &= \frac{18}{38} \cdot (1) + \frac{20}{38} \cdot (-1) \\ &= \frac{-2}{38} \\ &= \boxed{-0.05} \end{aligned}$$

Let's just do one more

Example 3

Toys are randomly placed in cereal boxes so that 1 out of every 6 boxes contain a toy. How many boxes **on average** must be bought to get a toy?

There are an infinite amount of cases

1. First box contains toy, $P(1) = \frac{1}{6}$
2. Second box contains toy, $P(2) = \frac{5}{6} \cdot \frac{1}{6}$
3. Third box contains toy, $P(3) = \left(\frac{5}{6}\right)^2 \cdot \frac{1}{6}$

4. You get the idea

We do $\frac{5}{6}$ because all the preceding boxes DO NOT contain toys, which has a $\frac{5}{6}$ chance.

Since the metric we are looking for is the number of boxes, the "payoff" is the number of boxes opened

$$\begin{aligned} \text{EV} &= \sum (\text{probability})(\text{payoff}) \\ &= 1 \cdot \frac{1}{6} + 2 \cdot \frac{5}{6} \cdot \frac{1}{6} + 3 \cdot \left(\frac{5}{6}\right)^2 \cdot \frac{1}{6} + 4 \cdot \left(\frac{5}{6}\right)^3 \cdot \frac{1}{6} + \dots \\ &= \sum_{k=1}^{\infty} k \cdot \left(\frac{5}{6}\right)^{k-1} \cdot \frac{1}{6} \end{aligned}$$

Recall 1

$$\sum_{k=0}^{\infty} k \cdot x^k = \frac{x}{(1-x)^2}, -1 < x < 1$$

I'm too lazy to write out the rest so

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